

ABDUL MANAN

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ABOUT ME

Software Engineer delivering scalable solutions across FinTech, Retail, AI, and Industrial Automation, with expertise in Full-Stack Web Development and DevOps. Skilled at working within multi-disciplinary teams and engaging directly with stakeholders, translating complex business needs into robust technical solutions for organisations across the Middle East and the United States.

EXPERIENCE

Software Engineer (Backend)

Sept 2025 - Current

Virtual World (IBEX)

Remote

- Designed & implemented a **Multi-Tenant Architecture** for a Ticketing System using Nest.js, PostgreSQL, and Keycloak, ensuring database isolation per tenant and enabling scalable client onboarding for major clients including FIA, Jazz, and Shifa.
- Optimized API performance by replacing sequential database calls with parallel execution and integrating data loaders to cache and batch related queries, **reducing queries by over 80%**.
- Built the entire **Workflows module** to enable ticket escalations to relevant stages, departments, and users while integrating multiple communication channels.
- Designed a custom **Audit Logging** system with a dedicated change log table to capture all data mutations, persisting before/after state snapshots along with actor metadata for full operational traceability.

Polyglot Engineer (Full Stack & DevOps)

June 2024 - July 2025

Cowlar Design Studio (YC-17)

Pakistan

- Engaged directly with clients to understand pain points, propose solutions, and refine them through iterative feedback. Led projects from 0-to-1, **owning the full Software Development Lifecycle** (SDLC) from requirements and UI design to team execution, client handoff, and post-launch support, all while serving as a dedicated Full-Stack and DevOps resource.
- Led the end-to-end development of a Subscription Management Project for a Qatari **FinTech** company, translating client needs into actionable goals and managing the execution of the development team.
- Optimized MongoDB queries, reducing query time by 75% on datasets with **17 million+ records**, enhancing data retrieval performance.
- Built a Gift Card Processing System, managing **QAR 4 million+** in transactions within 20 days of launch, ensuring system scalability and reliability.
- Designed & developed a Web Application using AWS Amplify, Lambda, and S3 to manage **thousands of simultaneous file uploads**, replacing a manual, error-prone process and improving visibility.
- Collaborated on developing annotation tools and integrating in-house and **Azure Machine Learning models**, streamlining the annotation process and cutting annotation time by over 50%.
- Worked on multiple products within our **Retail Ecosystem** such as E-Commerce Applications along with Customer and Admin Portals.
- Architected a **Bare-Metal Kubernetes Cluster** with MetalLB for load balancing, Calico for network policies, CRI-O as the container runtime, and Helm for app management, ensuring scalability and efficiency.
- Developed **CI/CD Pipelines** on GitLab/GitHub for seamless continuous integration and deployment.
- Leveraged **Node.js Worker Threads** to offload CPU-intensive tasks from the main event loop, improving application throughput and reducing response latency under high concurrency.
- Integrated various third-party features, including Email, WhatsApp and SMS along with **Apple, Samsung, and Google Wallet**.
- Implemented **MQTT protocol** for real-time IoT communication, using it as a broker for efficient data transmission between devices and applications.

- Built & maintained a **custom Role-Based Access Control** (RBAC) system to manage fine-grained user permissions and enhance system security.

Full Stack Developer
ezWifi (Techstars 23')

February 2024 - April 2024
Remote

- Developed an **analytics dashboard** to monitor and visualize key metrics for deployed ezRouters, delivering actionable insights to clients.
- Enhanced an **admin portal** to support remote router management, enabling administrators to perform configuration changes and restarts remotely.

Full Stack Developer
Freelancing

December 2023 - April 2024
Remote

- Stabilized **E-Commerce applications** by fixing bugs and developing new features as required.
- Worked with **Stripe API** to handle payments and subscriptions.

SKILLS

Frontend	Vue.js, React.js
Backend	Node.js, Nest.js, Express, Fastify, Python, FastAPI
Databases & ORMS	PostgreSQL, MongoDB, IndexedDB, InfluxDB, Sequelize, Prisma, MikroORM
CI/CD Tools	GitHub Actions, GitLab CI/CD
Cloud Platforms	Amazon Web Services (AWS)
DevOps	Git, Docker, Kubernetes, Helm
Others	GraphQL, Swagger, MQTT, Protocol Buffers, Linux, Bash Scripting
Languages	JavaScript, TypeScript, MongoDB Query Language, SQL

EDUCATION

Fast Nucs	Islamabad, Pakistan
Bachelor of Science in Computer Science	2020-2024
Supernova	Islamabad, Pakistan
Cambridge International A-Levels	2018-2020